MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Outotec
LARSEN & TOUBRO

Project: **Mansourah Massarah Gold Project**

Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**

49PR POWER PLANT STORAGE TANK AREA

DESIGN CALCULATION OF TANK FOUNDATION

AS BUILT

MD-216-49PR-EG-CV-CAL-2007

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Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

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REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria
MD-216-3000-EG-CV-RPT-1008	Geotechnical Report
SBC 301 :2007	Saudi Structural - Loading & Forces Code
SBC 304 :2018	Saudi Concrete Structures Code
API 650, 12th Edition	Welded Tanks For Oil Storage

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DEFINITIONS AND ABBREVIATIONS

Employer	MA'ADEN GOLD AND BASE METALS or their representatives
EPC Contractor	Consortium of Outotec (OT) and Larsen & Toubro (L&T)
Project management Company (PMC)	BECHTEL appointed by EMPLOYER
Supplier	An entity (supplier, consultant, contractor) under the direct project
Sub- supplier	A sub entity (supplier, consultant, contractor) under the direct project
Shall	Is to be understood as mandatory in relation to the requirements of this
Should	Is to be understood as a strong recommendation to comply with the
LRFD	Load and Resistance Factor Design
F.O.S	Factor of Safety

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1 INTENT

This document is prepared for design of HFO and LFO Storage (49PR) Tank Foundation. The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the foundation system.

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2 SCOPE

This document is prepared for design of Foundation for Storage Tanks in Power Plant Area. Design of foundations has been done in accordance with SBC-304, ACI-318M and API 650.

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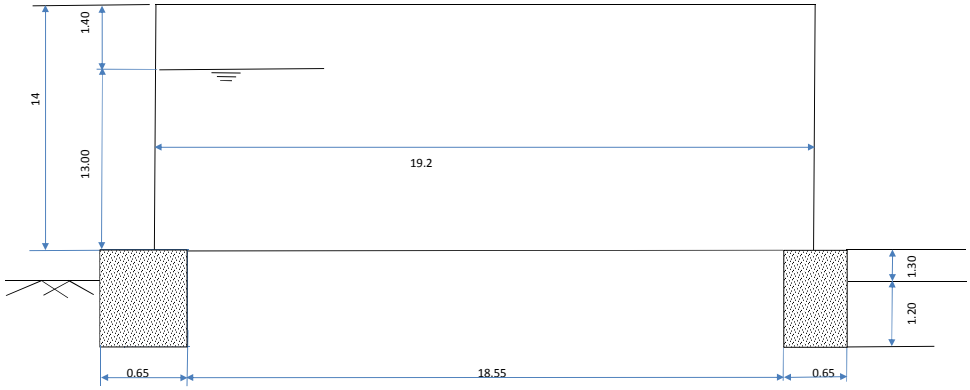
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3.0 Design of Foundations for HFO STORAGE TANK

(Foundation has been designed as a ring wall on the soil and it follows the principles stated in API 650)

INPUT FOR FOUNDATION: (REFER DOC. NO. DBAF490344 & DRAWING NO. MD-216-49PR-EG-ME-GAD-V553-1041)

Tank Loading Data				
	Loading Type			
HFO STORAGE TANK	Empyt (W)	1070	kN	
	Operating (F1)	35715	kN	3670*0.992*9.81
	Max (Wt)	38810	kN	3988*0.992*9.81



Summary of Tank Specification

Gross Volume of tank		=	3988 m^3	(REFER DRAWING NO. MD-216-49PR-EG-ME-GAD-V553-1041)
Net Volume of tank		=	3670 m3	
Diameter of Tank, D		=	19.2 m	
Height of Tank, H		=	14.4 m	
Height of Storage Oil, H_L		=	13 m	(Appox)
Inner Diameter of Concrete Ring Wall, ID		=	18.55 m	
Outer Diameter of Concrete Ring Wall, OD		=	19.85 m	
Height of Concrete Ring Wall, h		=	2.50 m	
Thickness of Concrete Ring Wall, t		=	0.65 m	
	P.C.D of Bolts	=	19.4 m	
Top of Foundation		=	1.3 m	(Above Finished Ground Level)
Unit Weight of Liquid, γ_L		=	9.92 kN/m^3	
Unit Weight of Soil, γ_S		=	20 kN/m^3	
Unit Weight of Concrete, γ_C		=	24 kN/m^3	

Wind Load & Seismic Load

Total Base Shear due to wind load	=	198.00 kN	(REFER DOC. NO. DBAF490344)
Total Base Shear due to Seismic load	=	1330.00 kN	

Load	**Load Condition**	**Base Shear (F,kN)**	**Moment (M,kN-m)**
Wind Loads	Normal	198.00	4932.00
Seismic Loads	Normal	1330.00	7199.00

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Foundation Weight

a) Ring Wall Weight = 2352.42 kN

a) Inside Soil Weight = 13512.87 kN

Sectional Area

Net cross-sectional area for empty condition, Ae = 39.21 m^2

Full cross-sectional area for operating condition, Ao= 309.46 m^2

Sectional Modulus

Sectional modulus for empty condition, Ze = 182.24 m^3

Sectional modulus for operating condition,Zo = 767.86 m^3

Check of Minimum Wall Thickness (As per API 650 ANNEX-B)

As the differential settlement is disadvantageous to reactor tank,minimum thickness of concrete wall required to equalize soil pressures between under reactortank and ringwall will be checked.

Soil pressure under the tank, $P = \gamma_L' H_L + \gamma_s' * h$

Soil pressure under the ringwall, $P = w / t + \gamma_L H_L / 2 + \gamma_c h$

Solving the above 2 equations, $\gamma_L' H_L + \gamma_s' * h = w / t + \gamma_L H_L / 2 + \gamma_c h$

or, $t = 2w / (\gamma_L' H_L - 2h(\gamma_c - \gamma_s))$

w = Weight of ring tank shell plus tributary weight of roof per meter circumference, $(W/\pi D)$ = 17.74 kN/m

So, Required Min Wall THK.= 0.3256 m = 325.6 mm < 650 mm OK

Basic Load Cases

1. Tank Dead Load & Platform load (D)
2. Operational Weight of slurry (F1)
3. Max Weight of slurry (F2)
4. Wind Load (W)
5. Seismic Load (E)

Allowable Stress Design

Load Combination (As per SBC-301 , Cl No. 2.4)

Term of Load	Load Conditions	Load Case	Load Combination	V (kN)	H (kN)	M (kNm)
Long term	Normal Operating	LC101 (2.4.1-1)	D + F1	36785.00	0	0
Short term	Empty	LC102 (2.4.1-7)	0.6 D + W	642.00	198.00	4932.00
	Operating	LC103 (2.4.1-3)	D + F1 + W	36785.00	198.00	4932.00
		LC104 (2.4.1-5)	D + F1+ 0.7E	36785.00	931.00	5039.30
		LC105 (2.4.1-8)	0.6 (D + F1) + 0.7E	22071.00	931.00	5039.30
	Maximun weight	LC106 (2.4.1-1)	D + F2	39880.00	0	0

Stability Check

Allowable Bearing Capacity of Soil, Q_{all} = 500 KN/m^2

An increase in allowable bearing capacities by about 33 percent may be considered under short term loading conditions like wind , earthquake.

Q_{all} = 665 KN/m^2

Load Case	Loads at bottom of foundation level		Qmax= V/A + M/Z kN/m^2	Q_{max}= V/A - M/Z kN/m^2	Q_{all}/ Q_{max}	Soil Bearing Check
	V (kN)	M(kNm)				
LC101	52650.30	0.00	170.13	170.13	2.94	OK
LC102	16507.30	5427.00	450.81	391.25	1.48	OK
LC103	52650.30	5427.00	177.20	163.07	3.75	OK
LC104	52650.30	7366.80	179.73	160.54	3.70	OK
LC105	37936.30	7366.80	132.18	112.99	5.03	OK
LC106	55745.30	0.00	180.13	180.13	2.78	OK

Sliding Check

Calculate safety factor against sliding by dividing resisting shear by actual shear

Fs =H resistance force / H acting force > 1.5

Load Case	Loads at bottom of V (kN)	Coef. of friction	H_{resist} (kN)	H_{acting} (kN)	Fs	Sliding Check
LC101	52650.30	0.4	21060.12	0.00	-	-
LC102	16507.30	0.4	6602.92	198.00	33.3	OK
LC103	52650.30	0.4	21060.12	198.00	106.4	OK
LC104	52650.30	0.4	21060.12	931.00	22.6	OK
LC105	37936.30	0.4	15174.52	931.00	16.3	OK
LC106	55745.30	0.4	22298.12	0.00	-	-

Overturning Check

Calculate safety factor against overturning by dividing resisting moment by overturning moment

The resultant Safety Factor should be 1.5 minimum

Fs =M resistance moment / M overturning moment > 1.5

Load Case	Loads at bottom of V (kN)	Radius of fdn.	M_{resist} (kNm)	M_{acting} (kNm)	Fs	Overturning Check
LC101	52650.30	9.925	522554.21	0.00	-	-
LC102	16507.30	9.925	163834.93	5427.00	30.2	OK
LC103	52650.30	9.925	522554.21	5427.00	96.3	OK
LC104	52650.30	9.925	522554.21	7366.80	70.9	OK
LC105	37936.30	9.925	376517.76	7366.80	51.1	OK
LC106	55745.30	9.925	553272.08	0.00	-	-

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Ultimate Strength Design

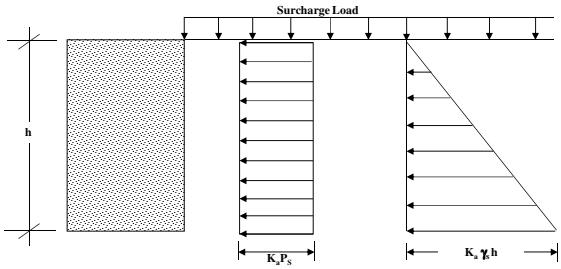
Load Combination (As per SBC-304 , Table 5.3.1)

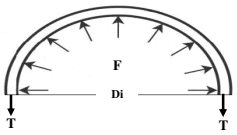
Term of Load	Load Conditions	Load Case	Load Combination	V (kN)	M (kNm)	A (m^2)	Z (m^3)
Long term	Normal Operating	LC101	1.4 (D + F1)	51499.00	0.00	289.53	694.87
Short term	Empty	LC102	0.9 D + 1.6*W	963.00	7891.20	289.53	694.87
	Operating	LC103	1.2 (D + F1) + 1.6*W	44142.00	7891.20	289.53	694.87
		LC104	1.2 (D + F1) + E	44142.00	7199.00	289.53	694.87
		LC105	0.9 (D + F1) + E	33106.50	7199.00	289.53	694.87
	Test	LC106	1.4 (D + F2)	55832.00	0.00	289.53	694.87

Surcharge Load

Load Case	Loads at bottom of Tank level		Surcharge= V/A + M/Z
	V (kN)	M(kNm)	kN/m^2
LC101	51499.00	0.00	177.87
LC102	963.00	7891.20	14.68
LC103	44142.00	7891.20	163.82
LC104	44142.00	7199.00	162.82
LC105	33106.50	7199.00	124.71
LC106	55832.00	0.00	192.84

Maximum Surcharge Load, Ps = 192.84 kN/m^2





Where, K_a = coefficient of earth pressure at rest = 0.5

P1 = $K_a * P_s$ = 96.42 KN/m^2

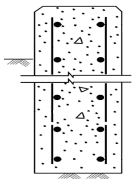
P2 = $1.4 * \gamma_S * h * K_a$ = 35.00 KN/m^2 for LC106 - 1.4 (D + F2)

Maximum Lateral Load

F = P1 * h + 1/2 * P2 * h = 284.80 KN/m

Hoop Tension Load

T= 1/2 * F * ID = 2641.49 KN



Hoop Tension Bar

1. Horizontal Reinfocement

ϕ = 0.9 (As per SBC 304 , Table No. 21.2.2)

Fy = 420 MPa

Required As = $T/(\phi \times Fy)$ = 6988.1 mm^2 (From hoop tension)

Minimum As = 0.0025 x t x h = 4062.5 mm^2 (As per API 650 , Cl. No. B.4.2.3)

Hence, Provide 30 Nos. D20= 8505.9 mm^2 OK

2. Vertical Reinfocement

Minimum As = 0.0015 * b * D = 2437.5 mm^2 (As per API 650 , Cl. No. B.4.2.3)

So, Used As = 2 row 5 bar D20 (@ 200mm)

=3142mm2 > Required As OK

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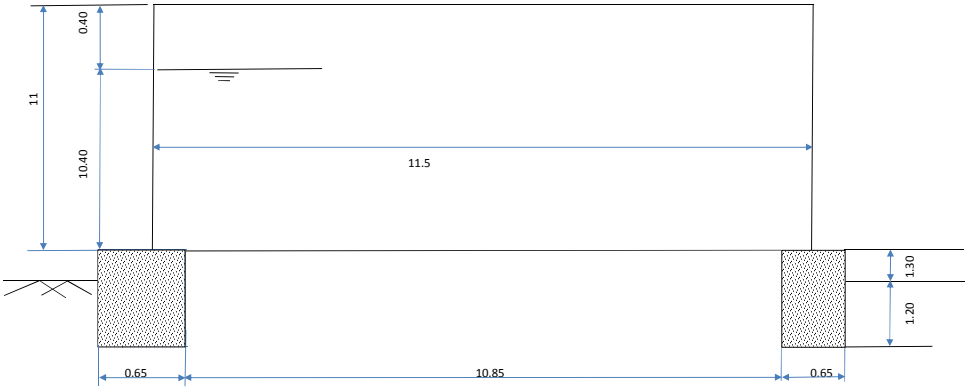
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4.0 Design of Foundation for LFO STORAGE TANK

(Foundation has been designed as a ring wall on the soil and it follows the principles stated in API 650)

INPUT FOR FOUNDATION: (REFER DRAWING NO. MD-216-49PR-EG-ME-GAD-V553-1042)

Tank Loading Data				
	Loading Type			
LFO STORAGE TANK	Empyt (W)	429	kN	
	Operating (F1)	8865	kN	1004*0.9*9.81
	Max (Wt)	9514	kN	1077.5*0.9*9.81



Summary of Tank Specification

Diameter of Tank, D		=	11.5 m	
Height of Tank, H		=	10.8 m	
Height of Storage Oil, H_L		=	10.4 m	(Appox)
Inner Diameter of Concrete Ring Wall, ID		=	10.85 m	
Outer Diameter of Concrete Ring Wall, OD		=	12.15 m	
Height of Concrete Ring Wall, h		=	2.50 m	
Thickness of Concrete Ring Wall, t		=	0.65 m	
	P.C.D of Bolts	=	11.7 m	
Top of Foundation		=	1.3 m	(Above Finished Ground Level)
Unit Weight of Liquid, γ_L		=	9 kN/m^3	
Unit Weight of Soil, γ_S		=	20 kN/m^3	
Unit Weight of Concrete, γ_C		=	24 kN/m^3	

Wind Load & Seismic Load

Total Base Shear due to wind load =	84.00 kN	(REFER DOC. NO. DBAF490344)
Total Base Shear due to Seismic load =	397.00 kN	

Load	**Load Condition**	**Base Shear (F,kN)**	**Moment (M,kN-m)**
Wind Loads	Normal	84.00	1196.00
Seismic Loads	Normal	397.00	1710.00

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Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 20-08-20 & REVISED AS MARKED	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Foundation Weight

a) Ring Wall Weight = 1409.00 kN

a) Inside Soil Weight = 4622.95 kN

Sectional Area

Net cross-sectional area for empty condition, Ae = 23.48 m^2

Full cross-sectional area for operating condition, Ao= 115.94 m^2

Sectional Modulus

Sectional modulus for empty condition, Ze = 64.11 m^3

Sectional modulus for operating condition,Zo = 176.09 m^3

Check of Minimum Wall Thickness (As per API 650 ANNEX-B)

As the differential settlement is disadvantageous to reactor tank,minimum thickness of concrete wall required to equalize soil pressures between under reactortank and ringwall will be checked.

Soil pressure under the tank, $P = \gamma_L' H_L + \gamma_s' * h$

Soil pressure under the ringwall, $P = w / t + \gamma_L' H_L / 2 + \gamma_c h$

Solving the above 2 equations, $\gamma_L' H_L + \gamma_s' * h = w / t + \gamma_L' H_L / 2 + \gamma_c h$

or, $t = 2 w / (\gamma_L' H_L - 2h(\gamma_c - \gamma_s))$

w = Weight of ring tank shell plus tributary weight of roof per meter circumference, $(W/\pi D)$ = 11.87 kN/m

So, Required Min Wall THK.= 0.3227 m = 322.7 mm < 650 mm OK

Basic Load Cases

1. Tank Dead Load & Platform load (D)
2. Operational Weight of slurry (F1)
3. Max Weight of slurry (F2)
4. Wind Load (W)
5. Seismic Load (E)

Allowable Stress Design

Load Combination (As per SBC-301 , Cl No. 2.4)

Term of Load	Load Conditions	Load Case	Load Combination	V (kN)	H (kN)	M (kNm)
Long term	Normal Operating	LC101 (2.4.1-1)	D + F1	9294.00	0	0
Short term	Empty	LC102 (2.4.1-7)	0.6 D + W	257.40	84.00	1196.00
	Operating	LC103 (2.4.1-3)	D + F1 + W	9294.00	84.00	1196.00
		LC104 (2.4.1-5)	D + F1+ 0.7E	9294.00	277.90	1197.00
		LC105 (2.4.1-8)	0.6 (D + F1) + 0.7E	5576.40	277.90	1197.00
	Maximun weight	LC106 (2.4.1-1)	D + F2	9943.00	0	0

Stability Check

Allowable Bearing Capacity of Soil, Q_{all} = 500 KN/m^2

An increase in allowable bearing capacities by about 33 percent may be considered under short term loading conditions like wind , earthquake.

Q_{all} = 665 KN/m^2

Load Case	Loads at bottom of foundation level		Qmax= V/A + M/Z kN/m^2	Q_{max}= V/A - M/Z kN/m^2	Q_{all}/ Q_{max}	Soil Bearing Check
	V (kN)	M(kNm)				
LC101	15325.96	0.00	132.19	132.19	3.78	OK
LC102	6289.36	1406.00	289.75	245.89	2.30	OK
LC103	15325.96	1406.00	140.17	124.20	4.74	OK
LC104	15325.96	1891.75	142.93	121.44	4.65	OK
LC105	11608.36	1891.75	110.86	89.38	6.00	OK
LC106	15974.96	0.00	137.78	137.78	3.63	OK

Sliding Check

Calculate safety factor against sliding by dividing resisting shear by actual shear

Fs =H resistance force / H acting force > 1.5

Load Case	Loads at bottom of V (kN)	Coef. of friction	H_{resist} (kN)	H_{acting} (kN)	Fs	Sliding Check
LC101	15325.96	0.4	6130.38	0.00	-	-
LC102	6289.36	0.4	2515.74	84.00	29.9	OK
LC103	15325.96	0.4	6130.38	84.00	73.0	OK
LC104	15325.96	0.4	6130.38	277.90	22.1	OK
LC105	11608.36	0.4	4643.34	277.90	16.7	OK
LC106	15974.96	0.4	6389.98	0.00	-	-

Overturning Check

Calculate safety factor against overturning by dividing resisting moment by overturning moment

The resultant Safety Factor should be 1.5 minimum

Fs =M resistance moment / M overturning moment > 1.5

Load Case	Loads at bottom of V (kN)	Radius of fdn.	M_{resist} (kNm)	M_{acting} (kNm)	Fs	Overturning Check
LC101	15325.96	6.075	93105.18	0.00	-	-
LC102	6289.36	6.075	38207.84	1406.00	27.2	OK
LC103	15325.96	6.075	93105.18	1406.00	66.2	OK
LC104	15325.96	6.075	93105.18	1891.75	49.2	OK
LC105	11608.36	6.075	70520.76	1891.75	37.3	OK
LC106	15974.96	6.075	97047.86	0.00	-	-

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
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Ultimate Strength Design

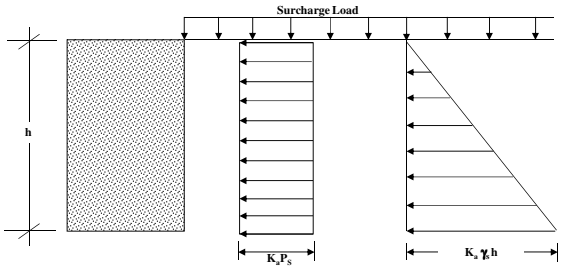
Load Combination (As per SBC-304 , Table 5.3.1)

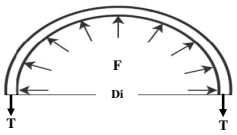
Term of Load	Load Conditions	Load Case	Load Combination	V (kN)	M (kNm)	A (m^2)	Z (m^3)
Long term	Normal Operating	LC101	1.4 (D + F1)	13011.60	0.00	103.87	149.31
Short term	Empty	LC102	0.9 D + 1.6*W	386.10	1913.60	103.87	149.31
	Operating	LC103	1.2 (D + F1) + 1.6*W	11152.80	1913.60	103.87	149.31
		LC104	1.2 (D + F1) + E	11152.80	1710.00	103.87	149.31
		LC105	0.9 (D + F1) + E	8364.60	1710.00	103.87	149.31
	Test	LC106	1.4 (D + F2)	13920.20	0.00	103.87	149.31

Surcharge Load

Load Case	Loads at bottom of Tank level		Surcharge= V/A + M/Z
	V (kN)	M(kNm)	kN/m^2
LC101	13011.60	0.00	125.27
LC102	386.10	1913.60	16.53
LC103	11152.80	1913.60	120.19
LC104	11152.80	1710.00	118.83
LC105	8364.60	1710.00	91.98
LC106	13920.20	0.00	134.02

Maximum Surcharge Load, Ps = 134.02 kN/m^2





Where, K_a = coefficient of earth pressure at rest = 0.5

P1 = $K_a * P_s$ = 67.01 KN/m^2

P2 = $1.4 * \gamma_S * h * K_a$ = 35.00 KN/m^2

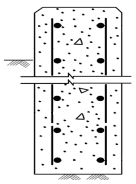
for LC106 | 1.4 (D + F2)

Maximum Lateral Load

F = P1 * h + 1/2 * P2 * h = 211.27 KN/m

Hoop Tension Load

T= 1/2 * F * ID = 1146.15 KN



Hoop Tension Bar

1. Horizontal Reinfocement

ϕ = 0.9 (As per SBC 304 , Table No. 21.2.2)

Fy = 420 MPa

Required As = $T/(\phi \times Fy)$ = 3032.1 mm^2 (From hoop tension)

Minimum As = 0.0025 x t x h = 4062.5 mm^2 (As per API 650 , Cl. No. B.4.2.3)

Hence, Provide 24 Nos. D20= 6804.7 mm^2 OK

2. Vertical Reinfocement

Minimum As = 0.0015 * b * D = 2437.5 mm^2 (As per API 650 , Cl. No. B.4.2.3)

So, Used As = 2 row 5 bar D20 (@ 200mm)

=3142mm2 > Required As OK

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Project Country	**KINGDOM OF SAUDI ARABIA**

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REFERENCE

[1] Structural Loading and Forces ; SBC Code 301: 2007
[2] Saudi Concrete Structures Code ; SBC 304 : 2018
[3] Civil,Structural and Architectural Design Criteria ; MD-216-1000-EG-ST-CRT-1001
[4] API 650 , 12th Edition
[5] Geotechnical Report No. MD-216-3000-EG-CV-RPT-1008

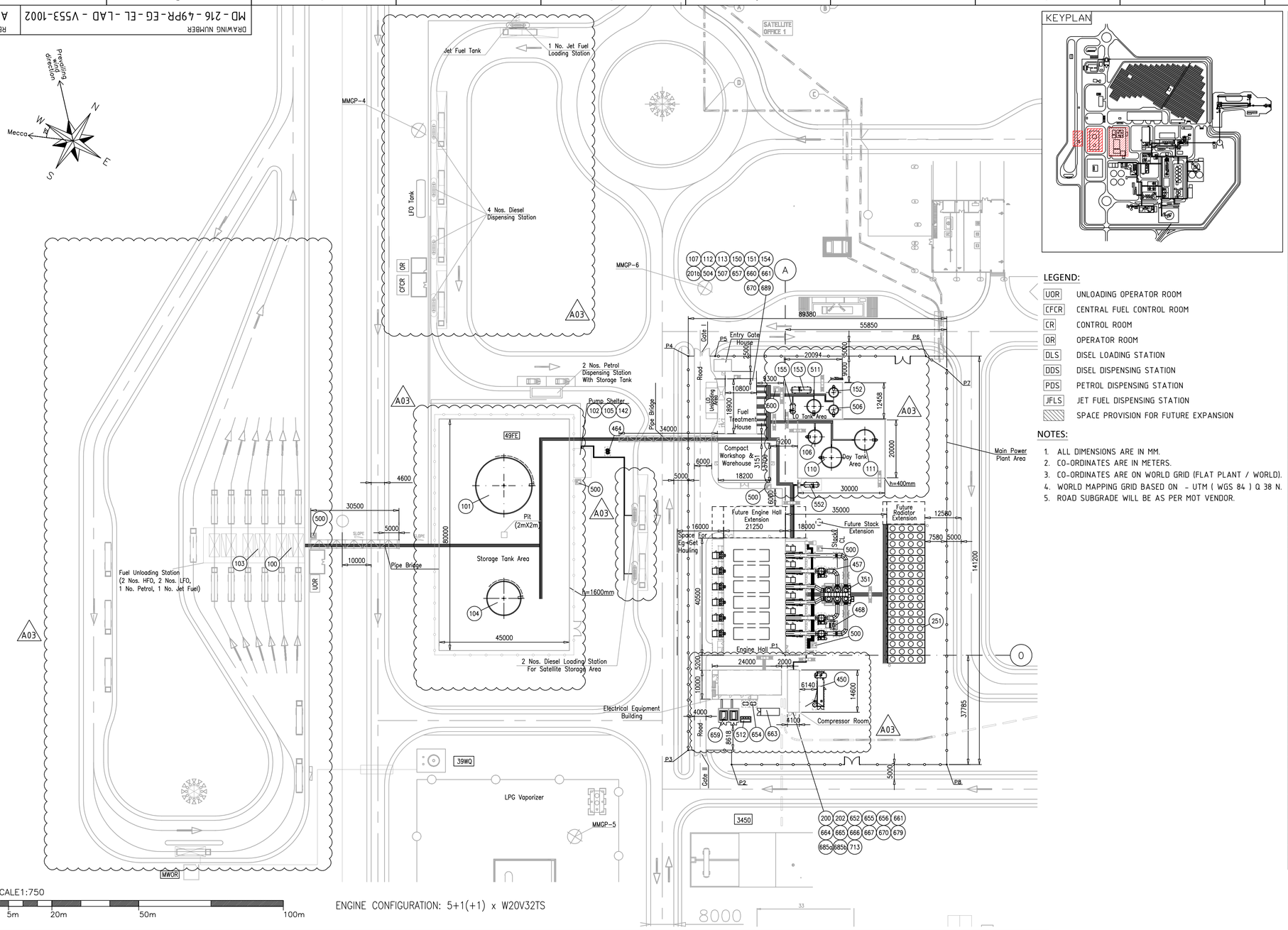
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ATTACHMENT 1 POWER GENERATION LAYOUT DRAWING

DRAWING NUMBER: MD - 216 - 49PR - EG - EL - LAD - V553-1002 | REV: A03



KEYPLAN

LEGEND:

UOR	UNLOADING OPERATOR ROOM
CFCR	CENTRAL FUEL CONTROL ROOM
CR	CONTROL ROOM
OR	OPERATOR ROOM
DLS	DISEL LOADING STATION
DDS	DISEL DISPENSING STATION
PDS	PETROL DISPENSING STATION
JFLS	JET FUEL DISPENSING STATION
	SPACE PROVISION FOR FUTURE EXPANSION

NOTES:

1. ALL DIMENSIONS ARE IN MM.
2. CO-ORDINATES ARE IN METERS.
3. CO-ORDINATES ARE ON WORLD GRID (FLAT PLANT / WORLD).
4. WORLD MAPPING GRID BASED ON - UTM (WGS 84) Q 38 N.
5. ROAD SUBGRADE WILL BE AS PER MOT VENDOR.

SITE AREA (This list applies to Engine Hall plan, Engine hall section and Site layouts)

Item No.	Pcs.	Code	Description	Volume [m³]	Weight [kg] Operative	Mounting level (Floor level)
251	18	VCA	Radiator (NL2)		6130	+5.000
351	6	NHA	Exhaust gas silencer		4790	+4.960
450	1	RHC	Heat recovery container		~19000	–
457	3	RCC	Exhaust gas boiler		17800	+0.221
464	1	RDB	Condensate tank		1350	–
468	1	DAD	Boiler washing water tank	5	~5620	–
500	6	DAA	Oily water collecting pit	2.5	–	–
512	1	–	Control pit		–	–
654	2	BAZ	Earthing transformer		–	–
659	2	BFB	Station transformer		–	–
663	1	BLM	Black starting unit		~5500	–

DAY / STORAGE TANK / UNLOADING AREA (This list applies to Power plant site layout)

Item No.	Pcs.	Code	Description	Volume [m³]	Weight [kg] Operative	Mounting level (Floor level)
100	2	PAA	HFO Unloading Pump (double)	–	1242	+0.200
101	1	PAB	HFO Storage Tank (frangible roof joint)	4000	4107000	+0.400
103	2	PAD	LFO Unloading Pump (double)		995	+0.200
104	1	PAE	LFO Storage Tank	1000	1042900	+0.400
106	1	PBA	HFO Buffer Tank	80	87800	+0.400
110	1	PBC	HFO Day Tank	150	163400	+0.500
111	1	PBF	LFO Day tank	150	163000	+0.400
152	1	QAC	LO Storage Tank (Clean)	35	39600	+0.400
153	1	QAD	LO Storage Tank (Used)	20	22590	+0.500
155	1	QAM	LO Service Tank	7	8090	+0.500
506	1	DDB	Sludge Tank	35	39700	+0.400
511	1	DAB	Oily Water Buffer Tank	50	56000	+0.400
552	1	VBC	Treated water tank	10	11450	–

FUEL TREATMENT HOUSE / PUMP SHELTER

Item No.	Pcs.	Code	Description	Volume [m³]	Weight [kg] Operative	Mounting level (Floor level)
102	1	PAC	HFO Transfer Pump Unit (double)		2645	±0.000
105	1	PAF	LFO Transfer Pump Unit (double)		486	±0.000
107	1	PBB	HFO Separator Unit (double)		4900	±0.000
112	1	PCA	HFO Feeder 3.0 (double)		3100	±0.000
113	1	PCA	LFO Feeder 3.0 (double)		1466	±0.000
142	2	–	LFO Loading Pump (double)		955	–
150	1	QAA	LO Unloading Pump Unit, (clean)		264	±0.000
151	1	QAB	LO Loading Pump Unit, (used)		264	±0.000
154	1	QAE	LO Transfer Pump Unit		665	±0.000
201b	1	TCC	Instrument air bottle	0.2	–	±0.000
504	1	DBB	Oily Water Treatment Unit		~11500	±0.000
507	1	–	Sludge Transfer Pump Unit		97	±0.000
657	1	BFA	LV Switchgear, Fuel treatment house		–	–
660	1	BJG	HFO Tank Contr Panel + Remote Commo		–	–
661	1	BLI	Lighting Panel *		–	–
670	3	–	Frequency Converters for Feeder Pumps		–	–
689	2	–	Frequency Converter for HFO Separator		–	–

ELECTRICAL EQUIPMENT BUILDING (This list applies to Engine Hall plan, Engine hall section and Site layouts.)

Item No.	Pcs.	Code	Description	Volume [m³]	Weight [kg] Incl. liquids	Mounting level (Floor level)
200	2	TCA	Instrument and working air unit		570	+2.100
202	2	TSA	Starting air unit		1200	+2.100
652	1	BAC	MV Switchgear			+2.100
655	2	BEY	DC-panel			+2.100
656	1	BFA	LV Switchgear			+2.100
661	1	BLI	Lighting panel			+2.100
664	1	BLN	Fire detection panel			+2.100
665	1	BLQ	Ventilation panel			+2.100
666	1	CFA	Control panel, common			+2.100
667	6	CFC	Control panel, engine wise			+2.100
670	6	BLP	Frequency converter, (Radiator)			+2.100
679	1		Desk for Phone/Printer/PC			+2.100
685a	1		Server rack		135	+2.100
685b	1		Communication cabinet			+2.100
713	3		Ventilation outlet fan			–

SITE INFO (This list applies to Engine Hall plan, section and Power lant site layouts)

FFL	±0.000 m	(+913.6 m from sea level)
FGL	−0.500 m	(+913.1 m from sea level)
RL	−0.300 m	(+913.3 m from sea level)

LEGEND (This list applies to Power plant site layout)

Colour:
- Items in customer scope
- Items in Wärtsilä scope

Linetype:
- Foundation/hidden structure
- Engine foundations
- Road center line
- Fence line
- Item above ground
- Site fence for main Power plant area
- Future extension

COORDINATES – UTM (WGS 84) Q38 N

No.	(E)x	(N)y	Description
P1	263987.528	2506005.175	A/0 point of Engine hall
P2	263991.831	2505963.310	Fence post
P3	263976.470	2505959.607	Fence post
P4	263904.291	2506075.108	Fence post
P5	263912.170	2506080.032	Fence post
P6	263973.959	2506118.642	Fence post
P7	263982.571	2506118.482	Fence post
P8	264054.918	2506002.731	Fence post

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A03	01/06/2020	REVISED AS PER COMMENTS DATED 13/05/2020	--	SJK	--	NBR	
A02	17/04/2020	REVISED AS PER COMMENTS DATED 16/03/2020	--	SJK	--	NBR	
A01	14/02/2020	ISSUED FOR APPROVAL	--	SJK	--	NBR	

QRS

DOCUMENT NUMBER | REFERENCE DOCUMENTS

WÄRTSILÄ

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Energy Business

DRAWING NUMBER	DBAF489643				REVISION	f
DATE	29-05-2020	DATE	29-05-2020	DATE	29-05-2020	
DRAWN	hslx01	CHECKED	mspx03	APPROVED	mnyx02	

LARSEN & TOUBRO DRAWING NUMBER: O19045_008_M_V_12021_001 | REVISION: 2

OUTOTEC DRAWING NUMBER: 900707 __GAU01 __WG01__MTL01 __45003 | DOCUMENT ID: OU100673503 | REVISION: 2

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DRAWN	--	--	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	SJK	12/02/2020	LAYOUT
ENGINEER	--	--	POWER GENERATION
APPROVED	NBR	14/02/2020	PLAN
EMPLOYER			
SCALE	1:750	A1	DRAWING NUMBER: MD - 216 - 49PR - EG - EL - LAD - V553-1002 | REV: A03

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
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ATTACHMENT 2 GA DRAWING OF HFO STORAGE TANK

MA'ADEN معادن

Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**

Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **GENERAL ARRANGEMENT DRAWING 49PR POWER GENERATION TANK - HFO STORAGE TANK**

Document Number: **MD-216-49PR-EG-ME-GAD-V553-1041**

Revision: **A01**

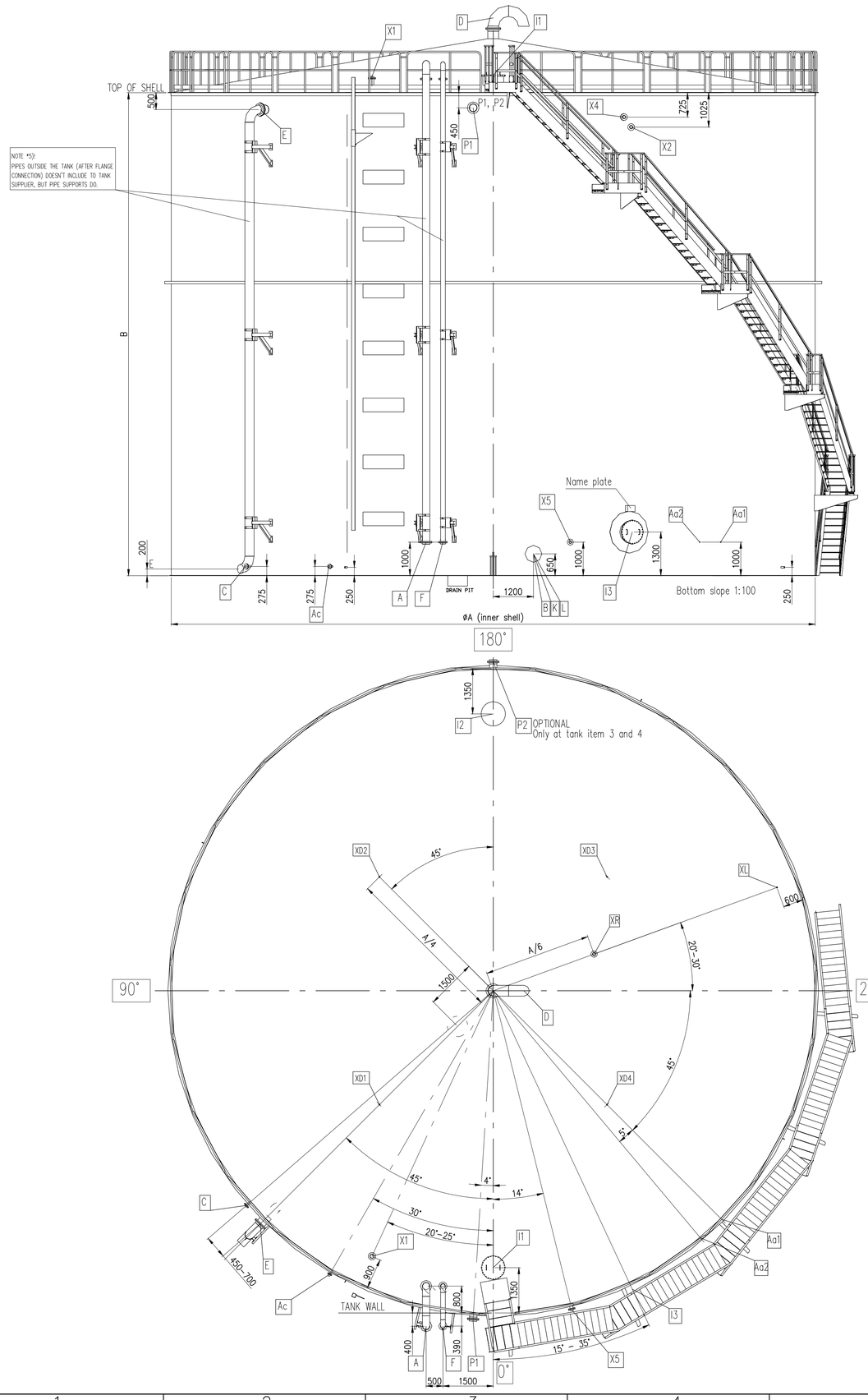
Revision Date: **01.04.2020**

Status: **ISSUED FOR INFORMATION**

Outotec Document Number:	900707	GAU01	WG01	MTL01	45031	OU100677312	0
L&T Document Number:	O19045	008	M	V	12162	001	0

WÄRTSILÄ Wärtsilä Finland Oy Energy Business	Wartsila Document Number :
	DBAA158867

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Rev	**Date**	**Revision Text**	**Prepared By**	**Reviewed By**	**Approved By**	**Employer**
A01	01.04.2020	FIRST ISSUE	Wartsila	SJK / DBH	NBR	



NOZZLE TABLE

MARK	DESCRIPTION	SIZE	FLANGE	ORIENTATION	PROJ. (min)	ELEVATION	REMARKS
A	FUEL OIL INLET	8" / DN200	X	2000 FROM 0°	150	ROOF	SEE NOTE *5)
B	FUEL OIL OUTLET, SEE SUCTION HEATER DRAWING		X	1200 FROM 0°		650	
C	DRAIN / EMPTYING	2" / DN50	X	450-700 FROM 45°	150	275	
D	VENTING PIPE / OPTIONAL FLAME ARRESTER	12" / DN300	X	AT CENTER	200	ROOF	WIRE MESH
E	OVERFLOW	10" / DN250	X	45°	225	500 FROM T.O.S.	SEE NOTE *5)
F	RETURN LINE	5" / DN125	X	1500 FROM 0°	150	ROOF	SEE NOTE *5)
K	STEAM INLET, SEE SUCTION HEATER DRAWING			1200 FROM 0°		650	
L	CONDENSE OUTLET, SEE SUCT. HEATER DRAWING			1200 FROM 0°		650	
I1	MANHOLE / EMERGENCY VALVE	24" / DN600	API 650	0°	250	ROOF	
I2	MANHOLE / EMERGENCY VALVE	24" / DN600	API 650	180°	250	ROOF	
I3	MANHOLE	24" / DN600	API 650	325°-345°	150	1300 WITH 1800MM PLATE WIDTH	CHECK THE LOCATION WITH DIFFERENT PLATE WIDTH
P1	FOAM POURER (OPTIONAL / BLIND FLANGE)	8" / DN200	X	4°	200	450 FROM T.O.S.	FOR 4000M3 - 10000M3
P2	FOAM POURER (OPTIONAL / BLIND FLANGE)	8" / DN200	X	180°	200	450 FROM T.O.S.	FOR 7500M3 - 10000M3
X2	HIGH LEVEL CONTROL	3" (150 lbs)	X	1800-2100 ABOVE STAIRS	175	1025 FROM T.O.S.	
X4	HIGH HIGH LEVEL CONTROL	3" (150 lbs)	X	1800-2100 ABOVE STAIRS	175	725 FROM T.O.S.	
X5	LOW LOW LEVEL CONTROL (BLIND FLANGE)	3" (150 lbs)	X	346°	175	1000	
Ac	CONNECTION FOR PRESSURE TRANSMITTER	1 1/2"(150 lbs)	X	30°	150	275	
X1	CONNECTION FOR LOCAL LEVEL INDICATOR		X	20°-25°	150	ROOF	
Aa1	TEMPERATURE LOCAL	G1/2"		315°		1000	
Aa2	TEMPERATURE SENSOR	G1/2"		320°		1000	
XR	ALT. CONN. FOR LEVEL MEASUREMENT(BLIND FLANGE)	3" (150 lbs)	X	240°-250°	150	ROOF	
XL	ALT. CONN. FOR HIGH HIGH-LEVEL CONTROL(PLUGGED)	G1"		240°-250°		ROOF	
XD1-XD4	HEAT DETECTORS (OPTIONAL / PLUGGED)	1/2" 14NPT		45°, 135°, 225°, 315°		ROOF	TOTAL 4 PCS
x ADDITIONAL INFO FROM API 650 DATA SHEETS							

ACCESSORIES

DESCRIPTION	SIZE	QTY.	REMARKS
ROOF HANDRAIL	H= 1070 mm	1 SET	
NAME PLATE	250 x 180	1 SET	ABOVE MANHOLE
STAIRS	-	1 SET	
EARTHING BRACKET	12 x 50 x 150	4 SETS	
PLATFORM		1 SET	
ANCHORING FOOTS			ACC TO PROJECT SPEC. TANKLIST
EMERGENCY VALVES ACC TO API 2000			x
INSULATION			x
HEATING COIL			x
DRAIN PIT	24" / DN600	1 SET	ACC. TO API, CENTER UOS.
x ADDITIONAL INFO FROM API 650 DATA SHEETS			

DESIGN CONDITIONS & MATERIALS
SEE TANK DATA SHEETS

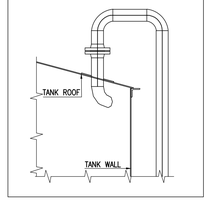
MEASURES				VOLUMES		PRELIMINARY DESIGN VALUES			
Item	Nominal size	ØA (mm)	B (mm)	Net (m3) *3)	Gross (m3) *4)	* Weight,Dry (kg)	Strength calc. with anch.	Strength calc. without anch.	Standard Data Sheet
1	4000m3 / 25200bbl	19200	14400	3669.4	3987.9	107000	WDAAA199444	WDAAA268542	DBAA165950
2	5000m3 / 31450bbl	21500	14400	4602.3	5001.7	127000	WDAAA199445	WDAAA268543	DBAA165947
3	7500m3 / 47200bbl	26300	14400	6886.3	7484.0	174000	WDAAA199446	WDAAA268544	DBAA165945
4	10000m3 / 62900bbl	30300	14400	9140	9933.2	224000	WDAAA199447	WDAAA268546	DBAA165943

Plates (mm)	4000m3	5000m3	7500m3	10000m3
Roof plate	8 **	8 **	8 **	8 **
Course 6-8	6 **	6 **	6 **	6 **
Course 5	6 **	6 **	6 **	8 **
Course 4	6 **	6 **	8 **	8 **
Course 3	8 **	8 **	10 **	10 **
Course 2	8 **	10 **	10 **	12 **
Course 1	10 **	10 **	12 **	14 **
Bottom plate	8 **	8 **	8 **	8 **

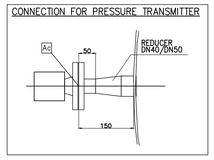
** Including corrosion allowance 1 mm (2mm for bottom plate)
*3) Between X5 and X4
*4) From tank bottom to lower part of E-connection

NOTE:
- API DATA SHEET TO BE FILLED IN EVERY CASE.
- FOR THE CIVIL DESIGN INPUT WEIGHT FROM PRELIMINARY STRENGTH CALCULATIONS.
- ALL DIMENSIONS ARE IN MILLIMETER.
- COUNTER FLANGES INCLUDED (WELDNECK).
- *FINAL WEIGHT OF THE TANK AND SPECIFIED STRENGTH CALCULATIONS BY TANK MANUFACTURER.
- SURFACE TREATMENT SEE DOCUMENT WDAAA203188 LATEST REVISION
- TANKS 4000m3 - 10000M3 => FRANGIBLE ROOF JOINT
- TANK TESTING ACC. TO API 650 (LATEST VERSION)
- WHEN SEISMIC ZONE IS 4, CHECK REQUIRED FREEBOARD (DISTANCE BETWEEN HH AND T.O.S.) FROM TANK MATRIX

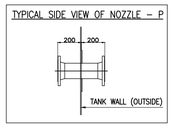
TYPICAL SIDE VIEW OF NOZZLE - A & F



CONNECTION FOR PRESSURE TRANSMITTER



TYPICAL SIDE VIEW OF NOZZLE - P



MA'ADEN DRAWING NUMBER :
MD-216-49PR-EG-ME-GAD-V553-1041

REV.	DATE	MADE	APPVD.	MEMO NO.	EXPLANATION
g	16-Nov-2017	ukax01	mri005		Freeboard note text added
f	20-Jan-2016	ukax01	mri005		X2, X4, P & E changed
e	16-Jun-2015	ukax01	mri005		XD1-4 added, P-connections changed

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Product: ACA ASS'Y DRG:

Dimensional drawing
HFO Storage Tank
4000m3 - 10000m3 ACC. TO API 650.
WF1-P STD
Proj. no. -

MADE	24-May-2007	Vesa Koskela	Units: mm/kg
CHKD	27-Jun-2007	Kenneth Edsvik	Page No. 1
APPD	27-Jun-2007	Marko Ristimäki	

Based on Document: Customer Document:

Scale: 1:75 Size: A1 Doc. ID: DBAA158867 g Rev.

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Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	A02	C01	D01
Date	30-07-2020	24-08-2020	19-09-2020	26-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 20-08-20 & REVISED AS MARKED	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
Date	30-07-2020	24-08-2020	19-09-2020	26-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 20-08-20 & REVISED AS MARKED	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

ATTACHMENT 3 GA DRAWING OF LFO STORAGE TANK

MA'ADEN معادن

Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project Mansourah Massarah Gold Project

Customer: MA'ADEN GOLD AND BASE METALS (MA'ADEN)

Title: GENERAL ARRANGEMENT DRAWING 49PR POWER GENERATION TANK - LFO STORAGE TANK

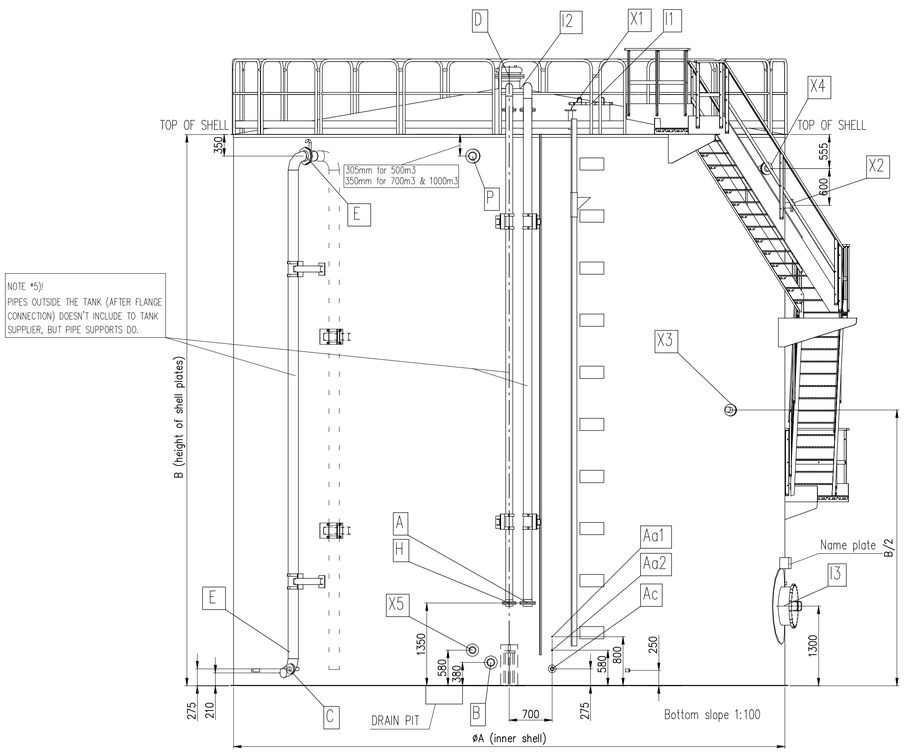
Document Number: MD-216-49PR-EG-ME-GAD-V553-1042
Revision: A01
Revision Date: 01.04.2020
Status: ISSUED FOR INFORMATION

Outotec Document Number:	900707	GAU01	WG01	MTL01	45032	OU100677313	0
L&T Document Number:	O19045	008	M	V	12163	001	0

WÄRTSILÄ Wärtsilä Finland Oy Energy Business	Wartsila Document Number :
	DBAA159441

REVISION HISTORY

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer
A01	01.04.2020	FIRST ISSUE	Wartsila	SJK / DBH	NBR	



NOZZLE TABLE

MARK	DESCRIPTION	SIZE	FLANGE	ORIENTATION	PROJ. (min)	ELEVATION	REMARKS
A	LFO FILLING	5" / DN125	x	300 FROM 0°	150	ROOF	SEE NOTE *5)
B	SUCTION TO UNIT	4" / DN100	x	300 FROM 0°	175	380	
C	DRAIN / EMPTYING	2" / DN50	x	50°	150	275	
D	VENTING PIPE / FLAME ARRESTER	12" - 150lbs	x	AT CENTER	200	ROOF	
E	OVERFLOW	6" / DN150	x	45°	200	350 FROM T.O.S.	SEE DWG DBAD550703 & NOTE *5)
H	RETURN FROM UNIT	4" / DN100	x	0°	150	ROOF	SEE NOTE *5)
I1	ROOF MANHOLE / EMERGENCY VALVE	24" / DN600	API 650	335°	250	ROOF	
I2	ROOF MANHOLE / EMERGENCY VALVE	24" / DN600	API 650	180°	250	ROOF	
I3	SHELL MANHOLE	24" / DN600	API 650	270° - 290°	150	1300 WITH 1800MM PLATE WIDTH	CHECK THE LOCATION WITH DIFFERENT PLATE WIDTH
Ac	CONN.FOR PRESSURE TRANSM.	1 1/2"(150 lbs)	x	700 FROM 0°	150	275	
X1	CONNECTION FOR LOCAL LEVEL INDICATOR		x	1000 FROM 0°	150	ROOF	
Aa1	TEMPERATURE LOCAL	G1/2"		700 FROM 0°		800	
Aa2	TEMPERATURE SENSOR (OPTIONAL / PLUGGED)	G3/4"		700 FROM 0°		580	
P	FOAM POURER (OPTIONAL / BLIND FLANGE)	4" / DN100	x	7°	175	305 FROM T.O.S.	FOR 500M3
P	FOAM POURER (OPTIONAL / BLIND FLANGE)	6" / DN150	x	7°	200	350 FROM T.O.S.	FOR 700M3 AND 1000M3
X2	HIGH LEVEL CONTROL	3" (150 lbs)	x	275°	175	1155 FROM T.O.S.	
X3	LOW LEVEL CONTROL (BLIND FLANGE)	3" (150 lbs)	x	1.8-2.1M ABOVE STAIRS	175	B/2	
X4	HIGH HIGH LEVEL CONTROL	3" (150 lbs)	x	295°	175	555 FROM T.O.S.	
X5	LOW LOW LEVEL CONTROL (BLIND FLANGE)	3" (150 lbs)	x	600 FROM 0°	175	580	
XR	ALT. CONN. FOR LEVEL MEASUREMENT (BLIND FLANGE)	3" (150 lbs)	x	240°-250°	150	ROOF	FOR 500m3-700m3
XR	ALT. CONN. FOR LEVEL MEASUREMENT (BLIND FLANGE)	3" (150 lbs)	x	240°-250°	150	ROOF	FOR 1000m3
XL	ALT. CONN. FOR HIGH HIGH-LEVEL CONTROL (PLUGGED)	G1"		260°-280°		ROOF	FOR 500m3-700m3
XL	ALT. CONN. FOR HIGH HIGH-LEVEL CONTROL (PLUGGED)	G1"		240°-250°		ROOF	FOR 1000m3
XD1-XD4	HEAT DETECTORS (OPTIONAL / PLUGGED)	1/2" 14NPT"		45°, 135°, 225° AND 315°		ROOF	TOTAL 4 PCS

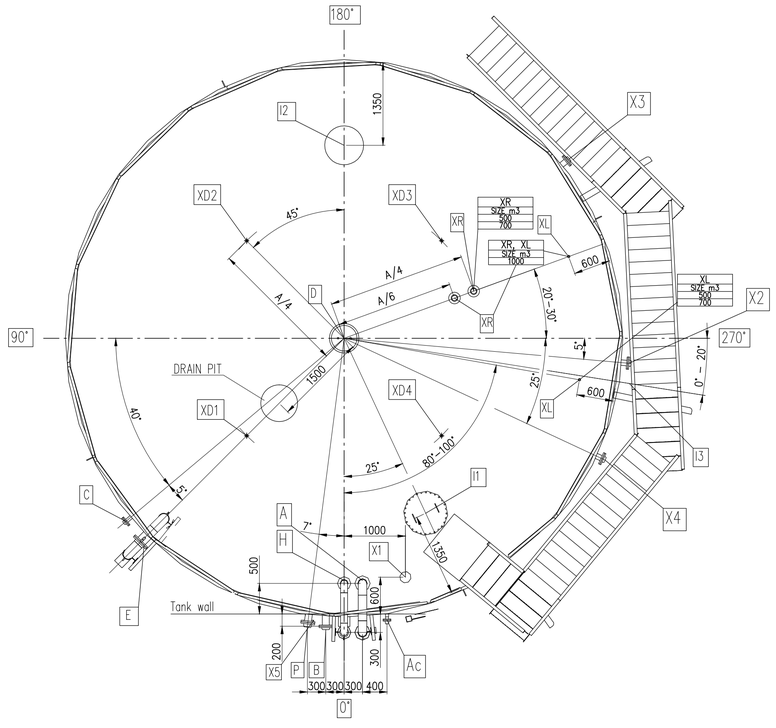
x ADDITIONAL INFO FROM API DATA SHEETS

ACCESSORIES

DESCRIPTION	SIZE	QTY.	REMARKS
ROOF HANDRAIL	H= 1070 mm	1 SET	
NAME PLATE	250 x 180	1 SET	ABOVE MANHOLE
STAIRS		1 SET	
EARTHING BRACKET	12 x 50 x 150	4 SETS	
PLATFORM		1 SET	
ANCHORING FOOTS			ACC TO PROJECT SPEC. TANKLIST
EMERGENCY VALVES ACC TO API 2000			x
INSULATION			x
HEATING COIL			x
DRAIN PIT	24" / DN600	1 SET	ACC. TO API, CENTER UOS.

x ADDITIONAL INFO FROM API DATA SHEETS

DESIGN CONDITIONS & MATERIALS
SEE TANK DATA SHEETS



MEASURES				VOLUMES		PRELIMINARY DESIGN VALUES			
Item	Nominal size	ø A (mm)	B (mm)	Net (m3) *3)	Gross (m3) *4)	* Weight,Dry (kg)	Strength calc. with anch.	Strength calc. without anch.	Standard Data Sheet
1	500m3 / 3150bbl	9000	9000	500.2	545.0	28300	WDAAA199437	WDAAA268535	DBAA167133
2	700m3 / 4410bbl	10500	9000	681.1	742.0	34800	WDAAA199438	WDAAA268536	DBAA167136
3	1000m3 / 6300bbl	11500	10800	1004.2	1077.4	42900	WDAAA199439	WDAAA268547	DBAA167138

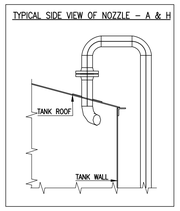
** Including corrosion allowance 1 mm (2mm for bottom plate)

*3) Between X5 and X4

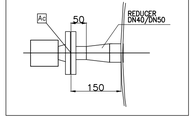
*4) From tank bottom to lower part of E-connection

Plates (mm)	500m3	700m3	1000m3
Roof plate	8 **	8 **	8 **
Course 6	-	-	6 **
Course 1-5	6 **	6 **	6 **
Bottom plate	8 **	8 **	8 **

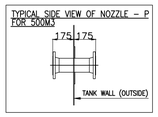
TYPICAL SIDE VIEW OF NOZZLE - A & H



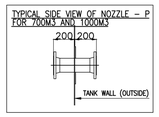
CONNECTION FOR PRESSURE TRANSMITTER



TYPICAL SIDE VIEW OF NOZZLE - P FOR 500M3



TYPICAL SIDE VIEW OF NOZZLE - P FOR 700M3 AND 1000M3



NOTE:

- API DATA SHEET TO BE FILLED IN EVERY CASE.
- FOR THE CIVIL DESIGN INPUT WEIGHT FROM PRELIMINARY STRENGTH CALCULATIONS.
- ALL DIMENSIONS ARE IN MILLIMETER.
- COUNTER FLANGES INCLUDED (WELDNECK).
- *FINAL WEIGHT OF THE TANK AND SPECIFIED STRENGTH CALCULATIONS BY TANK MANUFACTURER.
- SURFACE TREATMENT SEE DOCUMENT WDAAA203188 LATEST REVISION
- TANK TESTING ACCORDING TO API 650 (LATEST VERSION)
- LFO 500M3:
  -- WHEN SEISMIC ZONE IS 4, CHECK REQUIRED FREEBOARD (DISTANCE BETWEEN HH AND T.O.S.) FROM TANK MATRIX
- LFO 700-1000M3:
  -- WHEN SEISMIC ZONE IS 3 OR 4, CHECK REQUIRED FREEBOARD (DISTANCE BETWEEN HH AND T.O.S.) FROM TANK MATRIX

MA'ADEN DRAWING NUMBER :
MD-216-49PR-EG-ME-GAD-V553-1042

REV.	DATE	MADE	APPVD.	MEMO NO.	EXPLANATION
g	16-Nov-2017	ukax01	mri005		Freeboard note text added
f	20-Jan-2016	ukax01	mri005		P & D changed, XD1-4 added
e	01-Apr-2014	ukax01	mri005		Changes in I2, Aa1 & Aa2

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Power Plants

Product: | ACA | ASLY DRG:

MADE 25-May-2007 Kenneth Edsvik
CHKD 27-Jun-2007 vko004
APPD 27-Jun-2007 Marko Ristimäki

Units: mm/kg | Page No. 1

Dimensional drawing
LFO Storage Tank
500m3-1000m3. ACC. TO API650.
WFI-P STD

Proj. no. -

Based on Document: | Customer Document:

Scale: 1:50 | Size: A1 | Doc. ID: DBAA159441 g Rev.

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Confidential

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	A02	C01	D01
Date	30-07-2020	24-08-2020	19-09-2020	26-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 20-08-20 & REVISED AS MARKED	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
Date	30-07-2020	24-08-2020	19-09-2020	26-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 20-08-20 & REVISED AS MARKED	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

ATTACHMENT 4 LOAD TABLE FOR STORAGE TANKS DOCUMENT NO. DBAF490344

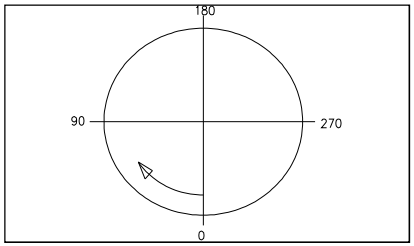
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Power Plants

Doc. Classification: Confidential

Title:	Tank list	**DocID:**	DBAF490344
		Revision:	f
Author:	Urpo Kähtävä	**Sheets:**	1
Approved by:	Shahu Injal / 28.07.2020	**Status:**	Approved
Project:	P/20002 Mansourah - Massarah Gold Mine		
Description:	Document and flow diagram numbers		
Type:	List		



Mansourah - Massarah Gold Mine, Saudi-Arabia, Design acc. to API 650, wind velocity 43m/s, Seismic Zone 1, design pressure 0.65kPa.
Horizontal tanks acc. to EN12285-2.

Tank	Nominal volume [m3]	Document number	Process Flow diagram	Diam.of tank [mm]	Height of shell [mm]	Qty	Net volume (betw. X4 and X5) [m3]	Number of anch.	Place of the first anchor bolt (°)	Diam. Of anch. bolts	Wind Shear Force (N)	Wind moment on tank (Nm)	Seismic Shear Force (N)	Ringwall overturning moment (Nm)	Slab Overturning moment (Nm)
HFO Storage Tank	4000m3	DBAF501378a	DBAF501831a	19200	14400	1	3662.5	24	0	M36	198,500	4,932,040	**1,330,100**	**7,199,480**	**10,490,000**
HFO Buffer Tank	80m3	DBAF501379b	DBAF504726a	4500	5400	1	68	6	0	M36	**41,045**	**116,870**	32,753	72,420	91,484
HFO Day Tank	150m3	DBAF501381c	DBAF504728a	6800	5400	1	155.2	8	0	M36	62,785	183,810	**64,801**	**133,520**	**191,050**
LFO Storage Tank	1000m3	DBAF499249a	DBAF504723a	11500	10800	1	993.5	14	0	M36	83,610	1,196,110	**396,700**	**1,710,690**	**2,259,100**
LFO Day Tank	150m3	DBAF499260b	DBAF504730a	6800	5400	1	155.9	8	0	M36	61,127	179,340	**64,209**	**131,980**	**189,470**
Lube Oil Storage Tank	35m3	DBAF501433b	DBAF502325a	3000	5400	1		6	0	M36	**25,795**	**72,275**	16,521	40,100	46,543
Lube Oil Service Tank	7m3	DBAF501457b	DBAF502380a	1600	length 3737	1		4	0	M20	-	-	-	-	-
Used Lube Oil Storage tank	20m3	DBAF501458b	DBAF504733a	2000	length 6959	1		4	-	M20	-	-	-	-	-
Sludge Tank	35m3	DBAF501382a	DBAF504737a	3000	5400	1		6	0	M36	**27,453**	**76,751**	16,534	38,965	45,217
Oily Water Buffer Tank	50m3	DBAF501459b	DBAF504735a	4500	3600	1		6	0	M36	**27,497**	**53,580**	20,036	27,722	38,559
Treated Water Storage Tank	10m3	DBAF501462b	DBAF504739a	1600	length 5347	1		4	-	M20	-	-	-	-	-

(DBAF454192-), Anchor bolt chair type A for tank size 300m3 and below.
(WDAAA272325b), Anchor bolt chair type B for tank size above 300m3.

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

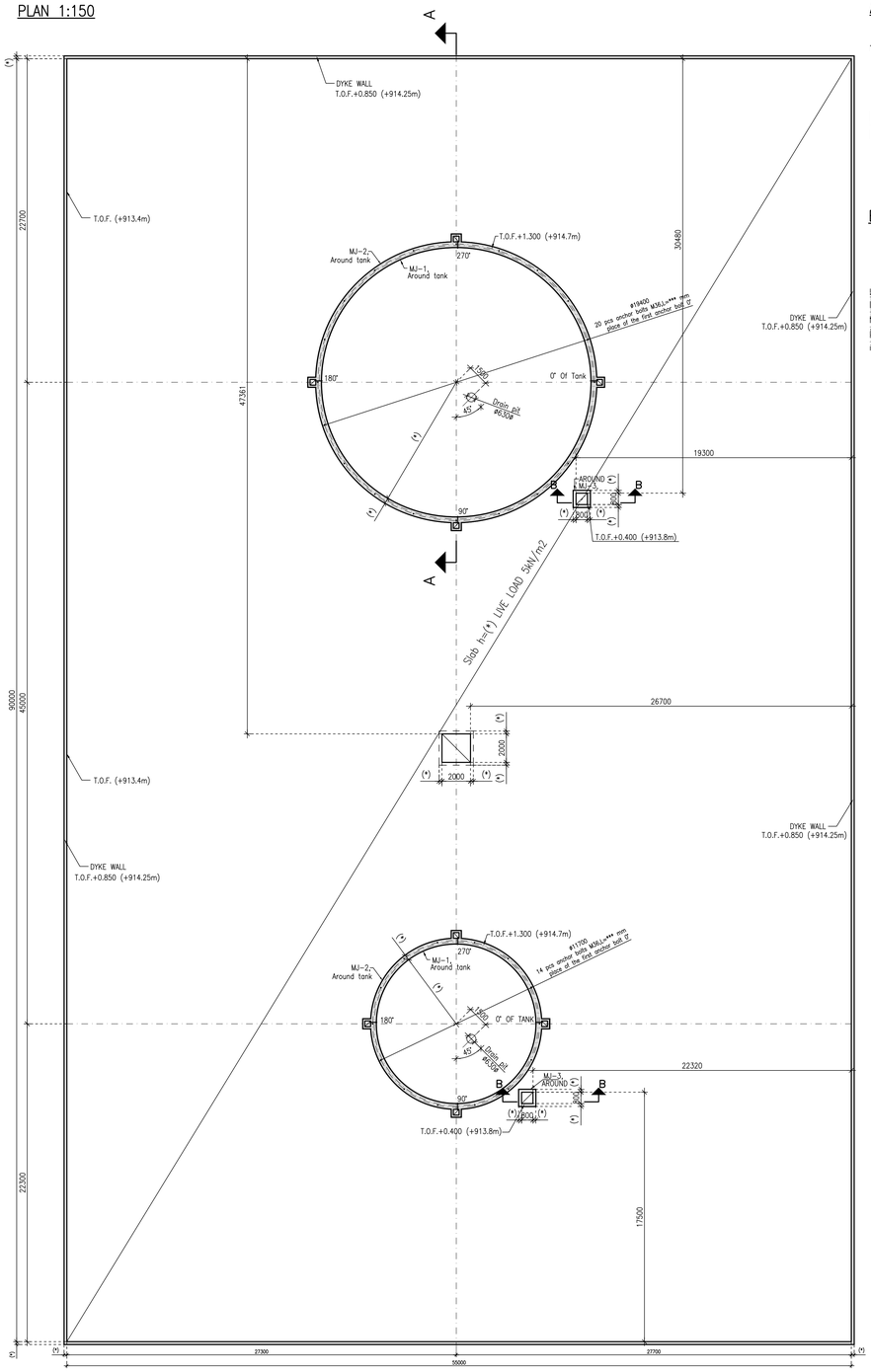
Ma'aden Revision No:	A01	A02	C01	D01
Date	30-07-2020	24-08-2020	19-09-2020	26-10-2023
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Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
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Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 20-08-20 & REVISED AS MARKED	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

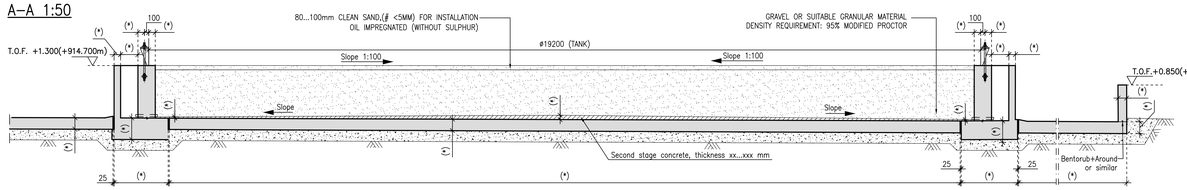
ATTACHMENT 5 STORAGE TANK AREA LAYOUT DRAWING

DRAWING NUMBER
MD - 216 - XXXX- XX- XX- XX - XXX - XXXX
REV A0X

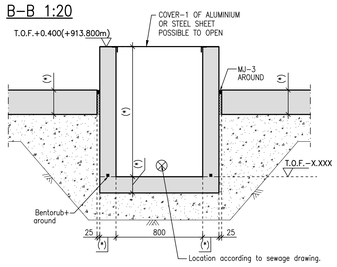
PLAN 1:150



A-A 1:50

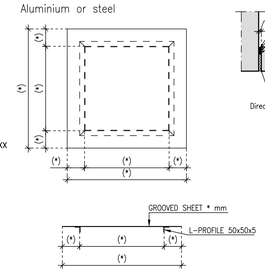


B-B 1:20

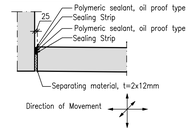


COVER-1 1:20

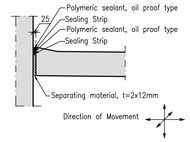
Aluminium or steel



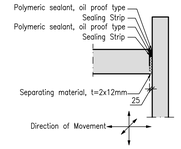
MJ-1 1:20



MJ-2 1:20



MJ-3 1:20



NOTE:
1) This drawing shall be mainly used for dimensional information purposes. Final construction drawing which concerns reinforcing and possible piling of the foundation to be done by the civil designer in accordance with the soil investigations at site. However below are given some points made by Wartsilg.
2) Beam size and location according to final design
3) MJ location according to final design
4) Conduit locations according to Cable Conduit design.
5) Drain points according to Unit Location drawing
6) Location and size of the drain channels to be checked according to final design

REFERENCE DRAWINGS	
DRG NO.	NAME OF THE DRAWING
DBAF518690	FOUNDATION LOCATION PLAN – BY CUSTOMER
DBAF490281	LAYOUT DRAWING STORAGE TANK AREA AND PUMP SHELTER
DBAF520574	ANCHOR BOLT CHAIR TYPE A, TANK SIZE 300m3 AND BELOW
DBAF520576	ANCHOR BOLT CHAIR TYPE B, TANK SIZE ABOVE 300m3
DBAF490344	TANK LIST DOCUMENT AND FLOW DIAGRAM NUMBERS
DBAF508433	LAYOUT DRAWING STORAGE TANK AREA – OILY WATER DRAINAGE
DBAF501378	DIMENSIONAL DRAWING HFO STORAGE TANK 4000m3
DBAF499249	DIMENSIONAL DRAWING LFO STORAGE TANK 1000m3
DBAF509368	LAYOUT DRAWING SITE WORKS 1/2 – UNDERGROUND PIPE SYSTEM
DBAF510203	SITE LOAD PLAN DRAWING

MATERIAL	AMOUNT
Concrete	*.* m3
Formwork	*.* m2
Bentorub+	*.* m
Second stage concrete	*.* m3
Polymeric sealant, oil proof type	*.* m
Sealing strip	*.* m
Separating material t=12 mm, h=150mm	*.* m
Separating material t=12 mm, h=225mm	*.* m
Separating material t=12 mm, h=350mm	*.* m
Separating material t=12 mm, h=700mm	*.* m
Inspection pipe: ∅50mm	*.* m
Waterstop JOCO D200 or similar	*.* m
L-profile 50x50x5	*.* m
Grooved steel sheet, t= * mm, ****x****	* pcs
HILTI HSA-M*x***	* pcs
UNP200	*.* m
Anchor bars ∅10	*.* m

TOLERANCE FOR LEVEL: TOP OF RINGWALL
(ACCORDING TO API 650)
TOP OF THE RINGWALL SHALL BE LEVEL WITHIN ± 1/8 INCH (3 mm) IN ANY 30 FEET (9000 mm) OF THE CIRCUMFERENCE AND ± 1/4 INCH (6 mm) IN THE TOTAL CIRCUMFERENCE MEASURED FROM THE AVERAGE ELEVATION.

ANCHOR BOLTS PROJECTION TOLERANCES SHALL BE +1/4" / –0" AND SHALL BE WITHIN 1/8" OF THEORETICAL LOCATION.

NOTES:
1. TOP OF FOUNDATIONS MARKED ACCORDING TO CUSTOMER'S INPUT
   T.O.F. +0.000 = (FFL) +913.400 m
2. " * "FOUNDATION SIZE AND LOCATION ACCORDING TO FINAL DESIGN.
3. THE BOLT EMBEDMENT LENGTH TO BE CALCULATED BY CIVIL (FOUNDATION) DESIGN.

FOR INFORMATION ONLY

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
	MD-216-1000-EG-ST-CRT-1001	A0X	XX/XX/XXXX	ISSUED FOR APPROVAL / REFERENCE / INFORMATION	---	MNP	---	QRS	

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
	CIVIL, STRUCTURAL & ARCHITECTURAL DESIGN CRITERIA	A0X	XX/XX/XXXX	ISSUED FOR APPROVAL / REFERENCE / INFORMATION	---	MNP	---	QRS	

WÄRTSILÄ

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Energy Business

DRAWING NUMBER: DBAF507559 — REVISION: a

DATE	15-Jul-2020	DATE	16-Jul-2020	DATE	17-Jul-2020
DRAWN	MPAX08	CHECKED	NCHX03	APPROVED	ANEX01

LARSEN & TOUBRO DRAWING NUMBER: G19045_XXX_X_X_XXXXX_XXX — REVISION: X
OUTOTEC DRAWING NUMBER: 900707 _ XXXXX_ XXXX_ XXXXX XXXXX — DOCUMENT ID: XXXXXXXXXXX — REVISION: X

LARSEN & TOUBRO

This drawing is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

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DRAWN	---	---
CHECKED	MNP	XX/XX/XXXX
ENGINEER	---	25/11/2019
APPROVED	QRS	XX/XX/XXXX
EMPLOYER		
SCALE	1:150	A0

MANSOURAH - MASSARAH GOLD PROJECT
Dimensional drawing
POWER GENERATION
STORAGE TANK AREA - INFORMATION DRAWING

DRAWING NUMBER: MD - 216 - XXXX- XX- XX- XXX - XXXX — REV: A0X